

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Even Semester Mid-Term Examination, 2025-26

Course Code: CEC 405
Course Name: Surveying

NITDGP/BTECH/Reg/Even/2026

Full Marks: 25
Time: 90 Minutes

Instructions: Answer all the questions.
Programmable scientific calculators are not allowed

Q	Body of the Question	Marks	Mapped CO															
1	a. A line was measured with steel tape which was exactly 30m at a pull of 5kg and the measured length was 229.62m. The pull applied during measurement was 10kg and the tape was uniformly supported. Find the true length of line if the cross-sectional area of tape was 0.02cm² and modulus of elasticity of tape material = 2.1X10⁶ kg/cm² b. Explain the role of base line, tie lines, check lines, and index sketch in ensuring accuracy.	3.5	CO2															
2	a. During a chain survey of NIT Durgapur campus, a 30 m chain was used. After completion of the survey, the chain was tested and found to be 30.12 m long. The measured lengths were: Base line AB = 285.60 m, Tie line AC = 162.40 m, Check line BC = 201.80 m Determine: i) The true lengths of AB, AC, and BC ii) The percentage error introduced in the measured area of the campus due to the incorrect chain length (Assume all measurements were taken using the same chain.)	2.5	CO1															
3	a. In a campus survey, a baseline PQ = 210 m was measured using a 20 m chain. Offsets to locate a laboratory building were taken as follows: - At 60 m from P, perpendicular offset = 28 m - At 140 m from P, perpendicular offset = 44 m - Later, the chain was found to be 20.10 m long. i) Calculate the measured area of the building plot ii) Determine the true area after applying the chain length correction iii) Comment on the effect of linear measurement errors on campus land-use planning	2.5	CO5															
3	a. The following observed bearings were recorded during a compass traverse conducted inside a campus: <table><thead><tr><th>Line</th><th>Fore Bearing</th><th>Back Bearing</th></tr></thead><tbody><tr><td>AB</td><td>46°30'</td><td>226°30'</td></tr><tr><td>BC</td><td>118°15'</td><td>298°45'</td></tr><tr><td>CD</td><td>202°00'</td><td>21°30'</td></tr><tr><td>DA</td><td>310°45'</td><td>130°45'</td></tr></tbody></table>	Line	Fore Bearing	Back Bearing	AB	46°30'	226°30'	BC	118°15'	298°45'	CD	202°00'	21°30'	DA	310°45'	130°45'	3	CO2
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Course Outcomes

- CO1: Learn basic principles of surveying and handling of various surveying instruments.
CO2: Learn to conduct engineering surveys.
CO3: Data entry in field books and level books.
CO4: Make and interpret maps.
CO5: Compute area and volumes

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i) Identify the stations affected by local attraction

ii) Determine the corrected bearings of all lines

iii) Calculate the included angles of the traverse

b. Write B.B for the followings bearings

i) $125^{\circ} 15'$ ii) $N30^{\circ} E$ iii) 360° iv) $S45^{\circ} 45' W$

2

CO2

4. List the equipment used in plane table surveying. Explain the function of each.

3

CO1

5. Study the contour map given in the figure below:

7

CO4

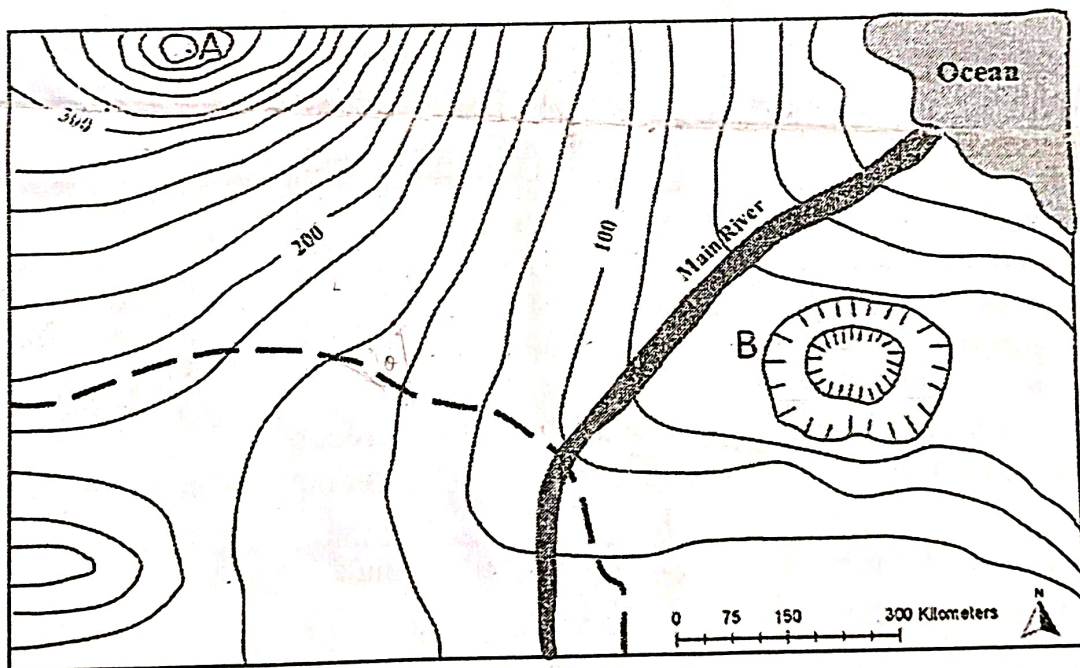
a) Determine the contour interval and explain how you obtained it

b) Determine the maximum possible elevation of Hill A.

c) Determine the minimum possible elevation of Depression B.

d) From the contour map, the river originates near the 200 m contour and meets the ocean at 0 m (sea level); if the measured length of the river between these two points on the map is 7 cm and the map scale is 1: 5,000,000, calculate the actual ground length of the river segment in kilometres.

e) Using the contour interval determined in part (a), calculate the gradient if the horizontal ground distance between two successive contour lines is 1 km



All unit in the map are in meter

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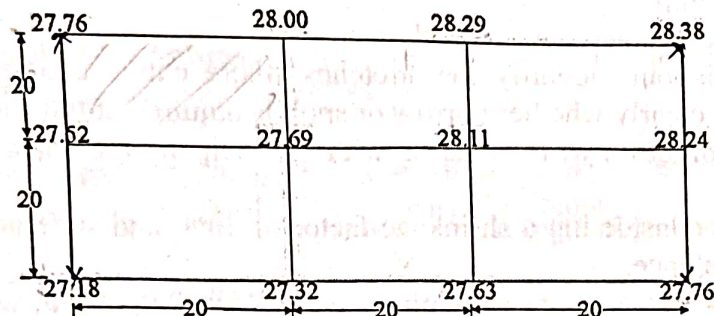
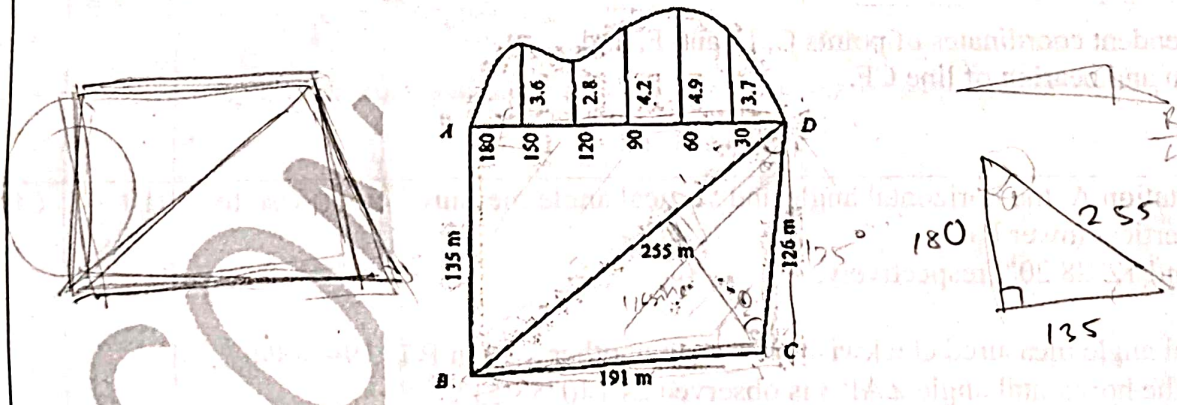
NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Even Semester End-Term Examination, 2025-26

NITDGP/BTECH/Reg/Even/2025-26

Course Code: CEC 405
Course Name: Surveying

Full Marks: 60
Time: 3 hours

Instructions: Answer all the questions.

Q	Body of the Question	Marks	Mapped CO																
1	a) A rectangular area is divided into a grid with 12 spot heights taken at equal spacing of 20 m, as shown in the figure. The ground is to be levelled to a horizontal plane at an elevation of 10.00 m. Calculate the volume of earthwork required using the spot height method  b) The same area is to be levelled such that the total cut is equal to the total fill. Assume no bulking or shrinkage of soil and neglect the effect of side slopes. Determine the formation level at which cut and fill are balanced.	5+5	CO5																
2	a) A piece of land has four sides. Three sides AB, BC and CD are straight lines, but the fourth side DA is irregular. The measured lengths are: AB = 135 m, BC = 191 m, CD = 126 m and BD = 255 m.  To measure the irregular side DA, offsets were taken outside the boundary. These offsets were measured at equal intervals of 30 m starting from point D. Determine the area of the land. <table><tr><td>Distance from D (m)</td><td>0</td><td>30</td><td>60</td><td>90</td><td>120</td><td>150</td><td>180</td></tr><tr><td>Offset (m)</td><td>0</td><td>3.7</td><td>4.9</td><td>4.2</td><td>2.8</td><td>3.6</td><td>0</td></tr></table>	Distance from D (m)	0	30	60	90	120	150	180	Offset (m)	0	3.7	4.9	4.2	2.8	3.6	0	7	CO5
Distance from D (m)	0	30	60	90	120	150	180												
Offset (m)	0	3.7	4.9	4.2	2.8	3.6	0												
2	b) With neat sketches, show the characteristic features of contour lines for the following: i) Hill ii) Vertical cliff (iii) Valley.	3	CO4																

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The volumes of earthwork between successive cross-sections at an interval of 100 m along a proposed road are given below							10	CO4
Chainage (m)	0-100	100-200	200-300	300-400	400-500	500-600		
Volume(m ³)	+200	+150	-100	-250	+300	-200		

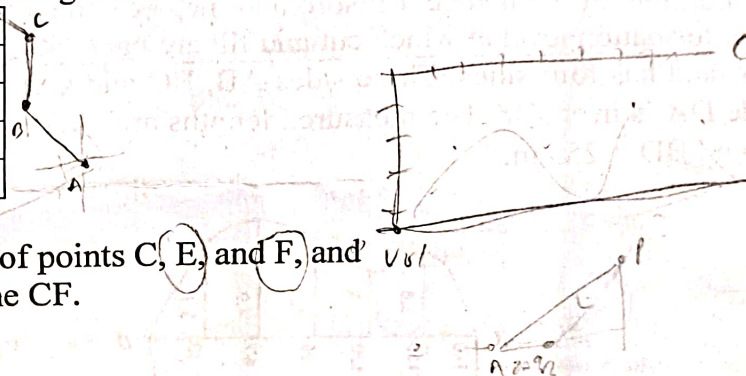
Note: Positive values indicate cut and negative values indicate fill.

Based on the above data:

- Draw the mass-haul diagram.
- Determine the balance points on the mass-haul diagram.
- Using the mass-haul diagram, identify the stretches where cut is used to fill adjacent sections. Also state clearly whether borrow or spoil is required at the end
- Indicate the direction of movement of earth with the help of the mass-haul diagram.
- Recalculate the volumes considering a shrinkage factor of 20%, and state how it affects the final earthwork balance.

A closed traverse A-B-C-D-E was run to determine the position of a point F, which lies midway between points A and E. The coordinates of point A are (500 m, 500 m). The lengths and whole-circle bearings of the traverse lines are given below:

Line	Length (m)	Bearing
AB	400	330°00'
BC	350	0°00'
CD	350	31°11'
DE	400	319°30'



Determine:

- the independent coordinates of points C, E, and F, and
- the length and bearing of line CF.

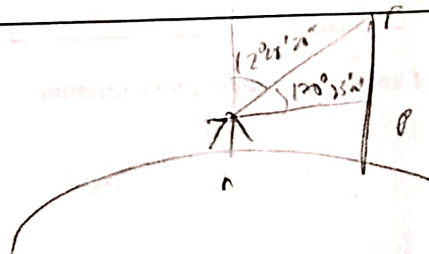
At a survey station A, the horizontal angle and vertical angle measured clockwise to the top of a vertical tower P are 170°25'30" and 12°28'20", respectively.

The horizontal angle measured clockwise from A to another station B is 199°48'40". At station B, the horizontal angle $\angle APB$ is observed as 140°53'23".

The horizontal distance between stations A and B is 264.10 m. The height of the instrument at station A is 1.30 m, and the height of the tower P is 40.80 m above ground level.

All angles are measured clockwise.

Determine the rise of the ground from station A to the base of the tower P.



6 A page from a levelling field book is shown below. Some of the staff readings are missing. The Reduced Level (RL) of the first benchmark BM1 is 150.000 m.

10

CO3

Using the Rise and Fall method:

1. Complete the levelling field book by determining the missing rise, fall, and reduced levels.
2. Perform the arithmetic check for the levelling work.
3. Determine the total rise and total fall.
4. If the known Reduced Level of the last point is 149.850 m, compute the closing error.
5. Assuming the closing error is to be distributed equally among all stations, determine the correction per station.

Station	Backsight (m)	Intersight (m)	Foresight (m)	Rise (m)	Fall (m)	Reduced Level (m)
BM1 ✓	2.745	—	—	—	—	150.000
TP1 ✓	—	2.310	—	?	?	?
TP2 ✓	—	2.685	—	?	?	?
TP3 ✓	—	—	2.125	?	?	?
TP4 ✓	2.560	—	—	?	?	?
TP5 ✓	—	2.420	—	?	?	?
TP6 ✓	—	2.950	—	?	?	?
TP7 ✓	—	—	2.305	?	?	?
TP8 ✓	2.880	—	—	?	?	?
Last Point	—	—	2.640	?	?	?

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1.915 ~ 2.04